

Ananya Deepak Deoghare

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SUMMARY

Senior ML Lead specializing in the architecture and scale of revenue-driving ranking systems, combining published research expertise with production-grade ML infrastructure to deliver multi-million dollar business impact.

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL
- **ML Systems:** Ranking Systems, Feature Stores, A/B Experimentation, Model Serving Pipelines, Optimization Algorithms
- **Frameworks:** PyTorch, TensorFlow, Scikit-learn
- **Infrastructure:** AWS, Hive, Redshift

EXPERIENCE

Algorithm Engineer

Jun 2023 – Present

QuinStreet, Foster City, CA

- Technical owner of end-to-end ML architecture for Personal Loans vertical (multi-million-dollar revenue line), responsible for ranking performance, deployment strategy, experimentation, and long-term system scalability.
- Designed scalable ranking pipeline serving 20–30K daily loan applicants (40% returning users), ensuring offline-online feature parity, reproducibility, and safe production rollouts.
- Designed and governed vertical-level feature store infrastructure, defining schema contracts, validation checks, and consistency of training to eliminate data leakage and reduce deployment risk.
- Increased revenue by 3–4% across publishers through production deployment of CTR prediction models; directly responsible for measurable P&L impact.
- Redesigned allocation logic using the Hungarian Algorithm for optimal assignment, replacing greedy heuristics and improving advertiser margin by 5–8%; improved robustness under constrained and edge-case scenarios.
- Designed a Cython high-performance simulation and scenario-testing engine to test stress-test ranking strategies before release, reducing runtime by 50% and enabling large-scale what-if analysis.
- Led experimentation roadmap and A/B testing framework design, defining primary metrics, guardrails, statistical validation, and rollback criteria for revenue-sensitive deployments.
- Built automated model monitoring, evaluation, and rollback pipelines to ensure stability across releases and alignment with long-term revenue KPIs.

Machine Learning Intern

Jun 2022 – Sep 2022

VidMob, New York, NY

- Built deep learning models for creative performance scoring, improving prediction accuracy by 25%.
- Integrated recommendation systems into production ad-tech pipelines.
- Designed model evaluation workflows improving engagement forecasting by 5%.

Student Researcher

Jul 2020 – Jan 2023

University of California, Los Angeles (UCLA)

- Co-authored peer-reviewed publication (SIGGRAPH 2022) on multimodal radar + RGB sensing for equitable remote photoplethysmography.
- Proposed shift-robust loss function reducing pulse prediction error by 40%.
- Built bias-mitigation ML pipelines improving fairness and accuracy by 75%.

EDUCATION

University of California, Los Angeles (UCLA)

2021 – 2023

M.S. Electrical and Computer Engineering

GPA: 3.97/4.0

PES University

2015 – 2019

B.Tech. Electronics and Communication Engineering

PUBLICATION

- **SIGGRAPH 2022** — Blending camera and 77 GHz radar sensing for equitable, robust plethysmography